Homework 5, Math 609

November 2, 2026

PROBLEM 1 (10 points):

- **(5 points)** Applying Newton's method for solving the equation $x^2 = a$, gives the following algorithm to compute the square root of a

$$t_{n+1} = \frac{1}{2}(t_n + \frac{a}{t_n})$$

  Find an algorithm for getting the cubic and quartic roots of a that have a similar form to the one above for the square root. Generalize your answer for finding the n-th root. Write a code to compute approximately $\sqrt{5}$. You will need to select the initial guess and the stopping criteria.

- **(5 points)** Consider the function $f(x) = x^3 - x + 1$. Find an appropriate interval and a function g so that the fixed point iteration method

$$x_{n+1} = g(x_n)$$

  will generate a sequence $\{x_n\}$ that will converge to the root of $f(x)$. Write a FPIM code to compute approximately the root. You will need to select the initial guess and the stopping criteria.

PROBLEM 2 (10 points): Let α be the solution of $f(x) = 0$, and $\{t_n\}$ be the sequence of approximate solutions, generated by the Newton's method. Show that

$$\alpha - t_{n+1} = -\frac{1}{2}\frac{f''(\xi_n)}{f'(t_n)}(\alpha - t_n)^2,$$

where ξ_n is between t_n and α.

PROBLEM 3 (10 points): The nonlinear system

$$x_1^2 - 10x_1 + x_2^2 + 8 = 0$$

$$x_1 x_2^2 + x_1 - 10x_2 + 8 = 0$$

can be transformed into the fixed point problem

$$x_1 = g_1(x_1, x_2) = \frac{x_1^2 + x_2^2 + 8}{10}$$

$$x_2 = g_2(x_1, x_2) = \frac{x_1 x_2^2 + x_1 + 8}{10}$$

- **(5 points)** Show that $G = (g_1, g_2)^T$, $G : [0, 1.5] \times [0, 1.5] \to \mathbb{R}^2$ has a unique fixed point in $[0, 1.5] \times [0, 1.5]$.
- **(5 points)** Write a code to compute approximately the root via the FPIM for systems. You will need to select the initial guess and the stopping criteria.

PROBLEM 4 (10 points): Derive the following two formulas for approximating derivatives and establish their error terms.

- **(5 points)**

$$f'(t) \approx \frac{1}{2h}\left[-3f(t) + 4f(t+h) - f(t+2h)\right]$$

- **(5 points)**

$$f''(t) \approx \frac{1}{12h^2}\left[-f(t+2h) + 16f(t+h) - 30f(t) + 16f(t-h) - f(t-2h)\right]$$

PROBLEM 5: (10 points) Consider the two point boundary value problem

$$-\phi'' + (4x^2 + 2)\phi = 0, \quad \phi(0) = 1, \quad \phi(1) = e,$$

with solution $\phi(x) = e^{x^2}$. Write a program to solve the problem using the second order finite difference scheme derived in class. Use step size

$$h = 2^{-p}, \quad p = 1, 2, \ldots, 14.$$

Solve the resulting linear system using

- LU factorization (Thomas algorithm)
- Gradient descent
- Conjugate gradient

For the iterative methods, you need to choose initial guess and stopping criteria. Turn in the code, plot the results and present the following table for each of the methods.

h	$\|u_h - \phi_h\|_\infty$	$\|u_h - \phi_h\|_\infty / h$	$\|u_h - \phi_h\|_\infty / h^2$	$\|u_h - \phi_h\|_\infty / h^3$